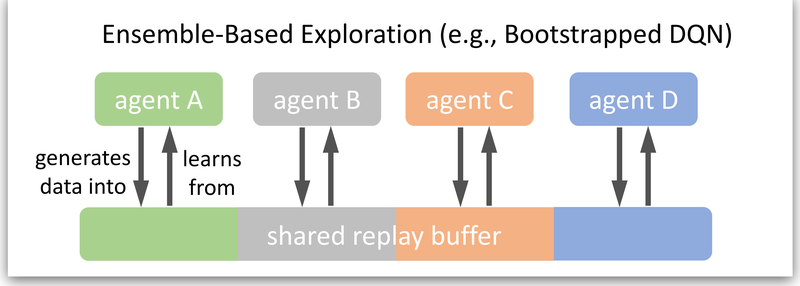
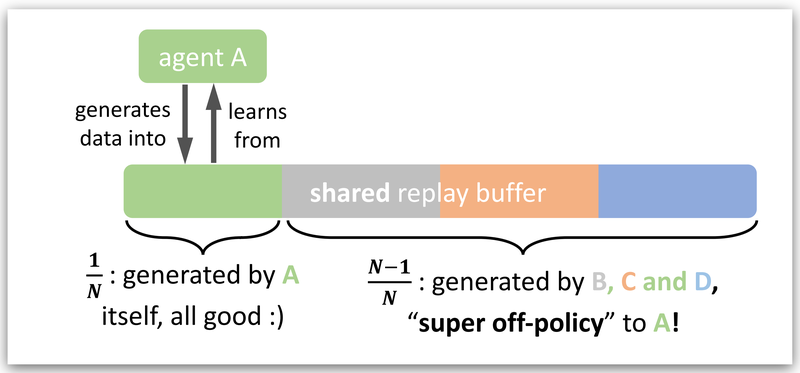
When training an ensemble of data-sharing agents, diversity can be a bad thing as it leads to challenging off-policy learning.

The Curse of Diversity in Ensemble-Based Exploration

Zhixuan Lin, Pierluca D'Oro, Evgenii Nikishin, Aaron Courville



Let's analyze the training data from agent A's perspective...

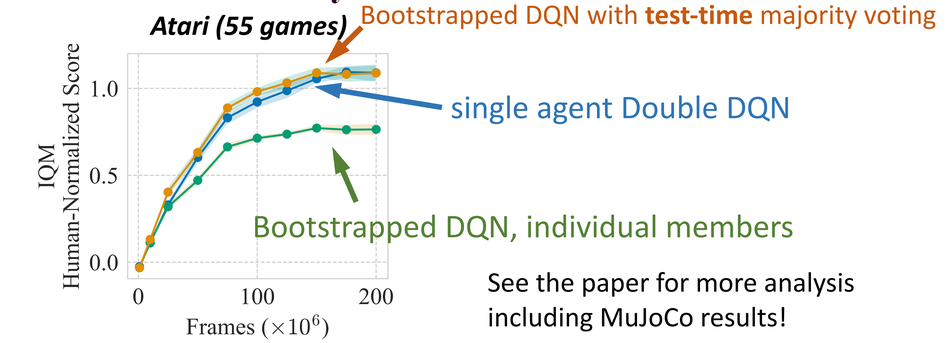


The Curse of Diversity: Individual members of a data-sharing ensemble can be much ***worse*** than an agent trained alone due to off-policy learning challenges in ensemble-based exploration.

Very challenging off-policy learning: similar to offline RL/tandem RL [1], but with a low proportion of online data.

[1] Ostrovski, G., Castro, P. S., & Dabney, W. (2021). The difficulty of passive learning in deep reinforcement learning. Advances in Neural Information Processing Systems, 34, 23283-23295.

The Curse of Diversity

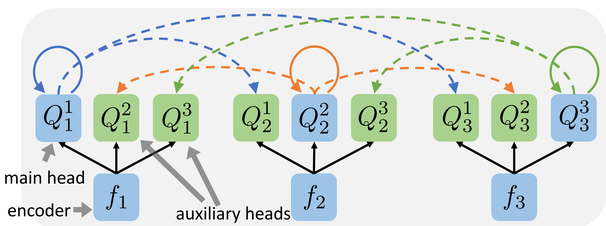


See the paper for more analysis including MuJoCo results!

Takeaways:

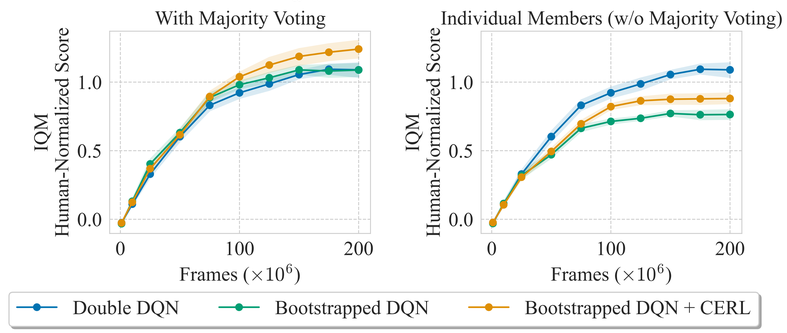
- **Curse of diversity:** Individual members of an ensemble can be much worse than an agent trained alone.
- Higher diversity (e.g., larger ensemble size) can make it even worse.
- Majority voting plays a much more important role than previously attributed (compared to deep exploration).

Cross-Ensemble Representation Learning (CERL)



- **Motivation**: tandem RL [1] shows good representations can help in a similar setting
- **Idea**: let different agents learn each other's value function as an auxiliary task

$$Q_i^j(s,a) \leftarrow r(s,a) + \gamma \max_{a'} \bar{Q}_j^j(s',a')$$



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full paper